Fiscal Year:	FY 2026	**Task Last Updated:**	FY 08/04/2025
PI Name:	Dev, Sheena Ph.D.		
Project Title:	Communication Delays in Cislunar Space: Behavioral Medicine Risk Characterization of Impacts on Private Family Conferences		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2023	**End Date:**	07/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date has changed to 07/31/2025 per L. Juliette/HFBP (Ed., 3/25/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bell, Suzanne Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
	This task will identify a research technology tool to enable controlled communications delays. This tool will then be used to create expected lunar communication time delays during simulated private family conferences (PFCs). The behavioral outcomes of communication delays on PFCs will be evaluated. The task will identify potential “breaking points” in delays associated with decrements in family and social connectedness and individual well-being. A controlled laboratory study with 27 family paired partners with close relationships will participate in a total of 5 sessions of simulated 30-minute audio with video PFCs. Communication delay conditions include 0, 2, 4, 6, and 8 seconds randomized across sessions. Each individual in the pair will complete surveys to evaluate their perceived connectedness with their paired partner. Project results will inform communication delay-durations and anticipated impacts for future research in NASA's Human Exploration Research Analog (HERA) Campaign 8. Aims:		

Task Description:	1. Identify NASA-approved technology to implement a communication delay at varying latencies during simulated private family conferences. 2. Identify the shortest communication delay latency (i.e., “break point") expected for Artemis missions and communication delay’s potential relation to decrements in family connectedness during simulated private family conferences. 3. Identify the shortest communication delay latency (i.e., “break point") expected for Artemis missions and communication delay’s potential relation to decrements in individual well-being during simulated private family conferences.
Rationale for HRP Directed Research:	This research is directed because NASA must define complete scientific activities in a short time and there is insufficient time to issue a solicitation.
Research Impact/Earth Benefits:	
Task Progress:	This entry reports on task progress from 08/2024 to 08/2025. During this time, the Behavioral Health & Performance (BHP) Laboratory completed all aspects of the study, including data collection, processing, and analysis for 35 study pairs undergoing 6 private family conference (PFC) sessions at delay latencies at 0, 4, 6, 8, 10, or 12 second one-way. A final summary report has been written and submitted to the NASA Human Factors and Behavioral Performance (HFBP) Element. The primary purpose of this final report was to summarize the results of the study and make recommendations for future spaceflight or analog missions. The "Communication Delays in Cislunar Space: Behavioral Medicine Risk Characterization of Impacts on Private Family Conferences. Final report” (PI: Dev) contains additional details. The final data has been transferred to NASA’s Life Sciences Data Archive (LSDA). Our main findings are summarized below: • PFCs improved relationship outcomes, relative to baseline, regardless of delay latency. • Relative to the real-time PFC, astronaut-like participants reported minimal group level decrements in relational closeness or relationship satisfaction after all delayed PFCs (Aim 2). They reported greater stress at 8 second one-way delays and longer, and greater difficulty concentrating at 10 second one-way delays and longer (Aim 3). • Relative to real-time, family rated lower relational closeness after 8 second one-way delays and reduced relationship satisfaction after 12 second one-way delays. They reported being more annoyed at 8 second delays and longer, and less cheerful at 12 second delays. • Both astronaut-like and family groups indicated all delay latencies significantly reduced their communication quality and would impact their relationships if they were required to continue communicating under those conditions for six months. Taken together, our results suggest PFCs at all delay latencies are likely to impact relationship outcomes in long-duration missions, with greater impacts to relationship and individual well-being after 8 second one-way delays. This preliminary data supports PFCs as a family support activity and offers the following recommendations for future spaceflight or analog missions: • Pre-mission training for family and crew to review strategies for effective communication with delays, expectations related to delayed communication, and protected time to practice prior to the mission. • Creative asynchronous communication methods should be offered to crew and their families for delays longer than 8 seconds one-way.
Bibliography Type:	Description: (Last Updated: 01/19/2026)
Articles in Peer-reviewed Journals	Dev SI, Moody ND, Arias DP, Bell ST. "Fly me to the Moon: Family conferencing with communication delays on lunar missions." Acta Astronaut. 2026 May;242:48-57. https://doi.org/10.1016/j.actaastro.2026.01.022 , May-2026